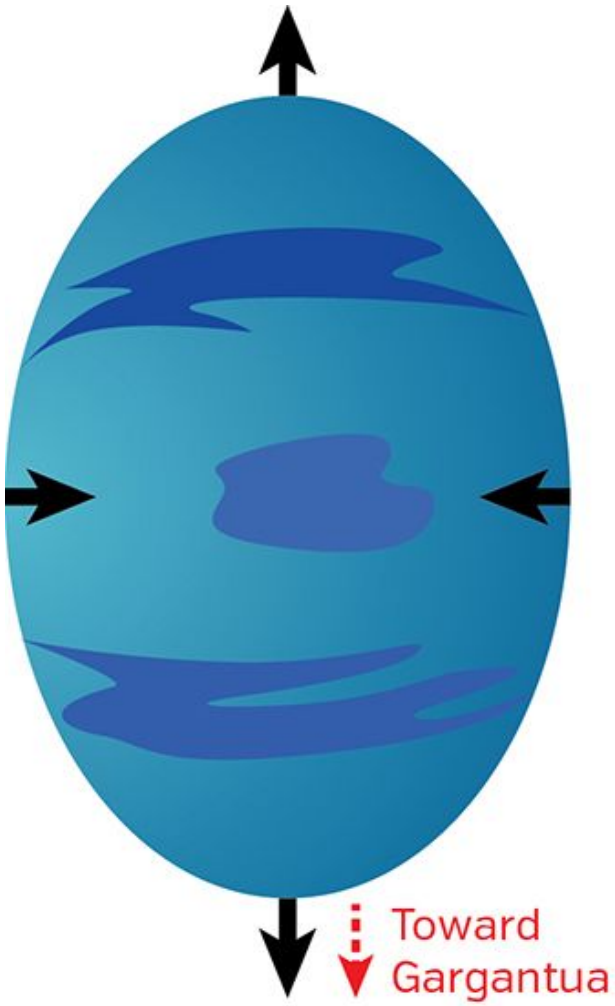


Fig. 6.1. Gargantua's tidal gravitational forces stretch and squeeze Miller's planet.

The strength of this stretch and squeeze is inversely proportional to the square of Gargantua's mass. Why? The greater Gargantua's mass, the greater its circumference, and therefore the more similar Gargantua's gravitational forces are on the various parts of the planet, which results in weaker tidal forces. (See Newton's viewpoint on tidal forces; Figure 4.8.) Working through the details, I conclude that Gargantua's mass must be at least 100 million times bigger than the Sun's mass. If Gargantua were less massive than that, it would tear Miller's planet apart!